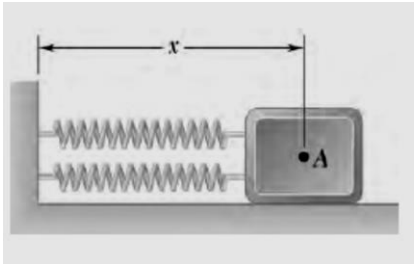
Mechanics **Tutorial Problems**

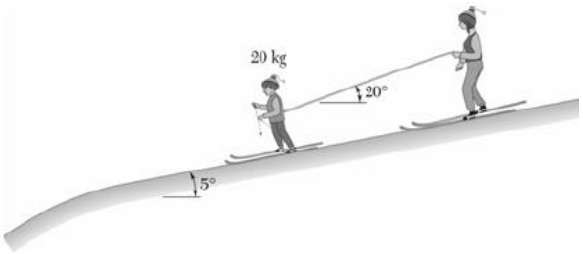
Date: 25 April, 2023

Q. 1: Point A oscillates with an acceleration a = 100(0.25 - x), where a and x are expressed in m/s^2 and meters, respectively. Knowing that the system starts at time t = 0 with v = 0 and x = 0.2 m, determine the position and the velocity of A when t = 0.2 s. (**Ans:** v = 1.921 m/s, x_m = 0.0981 m and 0.402 m)

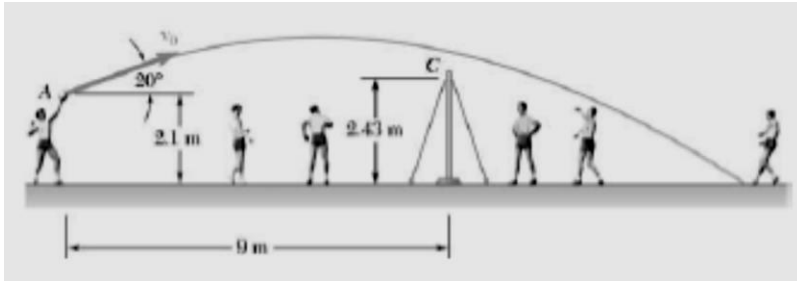


Q. 2: Based on observations, the speed of a jogger can be approximated by the relation $v = 7.5(1 - 0.04x)^{0.3}$, where *v* and *x* are expressed in mi/h and miles, respectively. Knowing that *x* = 0 at *t* = 0, determine (*a*) the distance the jogger has run when *t* =1 h, (*b*) the jogger's acceleration in ft/s^2 at *t* = 0, (*c*) the time required for the jogger to run 6 mi. (Ans: $x = 7.15$ km, $a = -52.1 \times 10^{-6}$ m/s^2, $t = 49.9$ min)

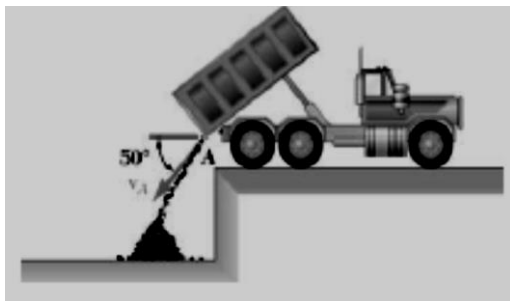
Q. 3: A mother and her child are skiing together, and the mother is holding the end of a rope tied to the child's waist. They are moving at a speed of 7.2 km/h on a gently sloping portion of the ski slope when the mother observes that they are approaching a steep descent. She pulls on the rope with an average force of 7 N. Knowing the coefficient of friction between the child and the ground is 0.1 and the angle of the rope does not change, determine (a) the time required for the child's speed to be cut in half, (b) the distance traveled in this time.



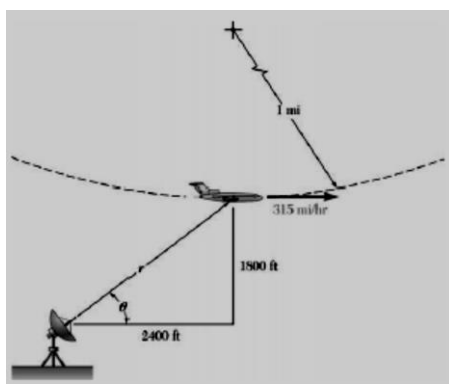
Q. 4: A volleyball player serves the ball with an initial velocity v_0 of magnitude 13.40 m/s at an angle of 20° with the horizontal. Determine (a) if the ball will clear the top of the net, (b) how far from the net the ball will land.



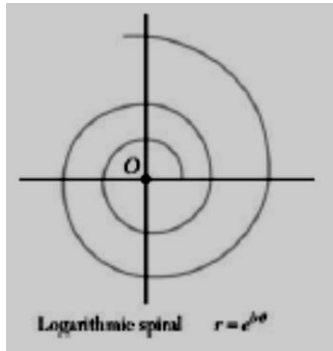
Q. 5: Coal is discharged from the tailgate *A* of a dump truck with an initial velocity $\mathbf{v}_A$ = 2 m/s at angle 50° as shown. Determine the radius of curvature of the trajectory described by the coal (*a*) at Point *A*, (*b*) at the point of the trajectory 1 m below Point *A*.



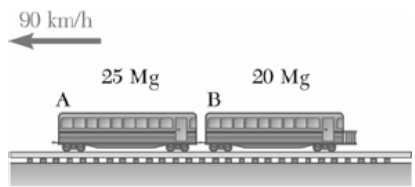
Q. 6: At the bottom of a loop in the vertical plane, an airplane has a horizontal velocity of 315 mi/h and is speeding up at a rate of 10 ft/s^2. The radius of curvature of the loop is 1 mi. The plane is being tracked by radar at O. What are the recorded values of $\dot{r}$, $\ddot{r}$, $\dot{\theta}$, $\ddot{\theta}$ for this instant?



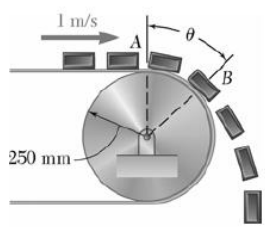
Q. 7: A particle moves along the spiral shown, Knowing that $\dot{\theta}$ is constant and denoting this constant by ω, determine the magnitude of the acceleration of the particle in terms of b, θ, and ω .

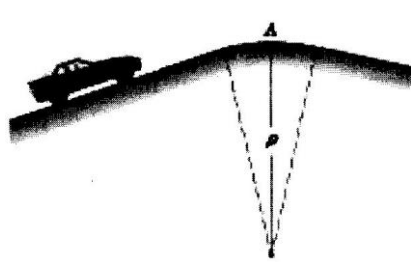


Q. 8: A light train made up of two cars is traveling at 90 km/h when the brakes are applied to both cars. Knowing that car A has a mass of 25 Mg and car B a mass of 20 Mg, and that the braking force is 30 kN on each car, determine (a) the distance traveled by the train before it comes to a stop, (b) the force in the coupling between the cars while the train is showing down.



Q. 9: A series of small packages, each with a mass of 0.5 kg, are discharged from a conveyor belt as shown. Knowing that the coefficient of static friction between each package and the conveyor belt is 0.4, determine (*a*) the force exerted by the belt on a package just after it has passed Point *A*, (*b*) the angle θ defining the Point *B* where the packages first *slip* relative to the belt.



Q. 10:

During a high-speed chase, an 1100-kg sports car traveling at a speed of 160 km/h just loses contact with the road as it reaches the crest *A* of a hill. (*a*) Determine the radius of curvature ρ of the vertical profile of the road at *A*. (*b*) Using the value of ρ found in part *a*, determine the force exerted on a 70-kg driver by the seat of his 1400-kg car as the car, traveling at a constant speed of 80 km/h, passes through *A*.